PREMIUM

Full-Length Mock Test

Mock 03

Mechanical Engineering

GATE ME

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | Moderate 33% | Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Engineering Mechanics — Kinetics]

A particle with constant velocity has acceleration:

A) Zero
B) Constant
C) Increasing
D) Decreasing

Q.2 [1 Mark] [EASY] [Strength of Materials — Bending]

Hooke's Law: stress proportional to:

A) Strain
B) Yield strength
C) Poisson's ratio
D) Density

Q.3 [1 Mark] [EASY] [Theory of Machines — Gears]

Reynolds number for laminar pipe flow:

A) Re < 1000
B) Re < 2000
C) Re < 4000
D) Re < 10000

Q.4 [1 Mark] [EASY] [Heat Transfer — Boiling]

Fourier's law negative sign indicates:

A) k is negative
B) Area negative
C) Heat flows opposite to temperature gradient
D) Heat flows along gradient

Q.5 [1 Mark] [EASY] [Thermodynamics — Entropy]

First law of thermodynamics is based on:

A) Mass conservation
B) Energy conservation
C) Momentum conservation
D) Zeroth law

Q.6 [1 Mark] [EASY] [Manufacturing — Forming]

In lathe, material is removed by:

A) Shear
B) Compression
C) Tension
D) Bending

Q.7 [1 Mark] [EASY] [Fluid Mechanics — Compressible Flow]

Torque transmitted by shaft is proportional to:

A) D

B) D^2

C) D^3

D) D t

Q.8 [1 Mark] [EASY] [Engineering Mechanics — Impact]

Carnot cycle consists of:

A) 2 isothermal + 2 adiabatic
B) 4 isothermal
C) 4 adiabatic
D) 2 isothermal + 2 isobaric

Q.9 [1 Mark] [EASY] [Strength of Materials — Thin Cylinders]

A particle with constant velocity has acceleration:

A) Zero
B) Constant
C) Increasing
D) Decreasing

Q.10 [1 Mark] [EASY] [Theory of Machines — Flywheels]

Hooke's Law: stress proportional to:

A) Strain
B) Yield strength
C) Poisson's ratio
D) Density

Q.11 [1 Mark] [EASY] [Heat Transfer — Convection]

Reynolds number for laminar pipe flow:

A) Re < 1000
B) Re < 2000
C) Re < 4000
D) Re < 10000

Q.12 [1 Mark] [EASY] [Thermodynamics — Entropy]

Fourier's law negative sign indicates:

A) k is negative
B) Area negative
C) Heat flows opposite to temperature gradient
D) Heat flows along gradient

Q.13 [2 Marks] [MODERATE] [Manufacturing — Metrology]

First law of thermodynamics is based on:

A) Mass conservation
B) Energy conservation
C) Momentum conservation
D) Zeroth law

Q.14 [2 Marks] [MODERATE] [Fluid Mechanics — Fluid Properties]

In lathe, material is removed by:

A) Shear

B) Compression

C) Tension

D) Bending

Q.15 [2 Marks] [MODERATE] [Engineering Mechanics — Impact]

Torque transmitted by shaft is proportional to:

A) D
B) D^2
C) D^3
D) D t

Q.16 [2 Marks] [MODERATE] [Strength of Materials — Torsion]

Carnot cycle consists of:

A) 2 isothermal + 2 adiabatic
B) 4 isothermal
C) 4 adiabatic
D) 2 isothermal + 2 isobaric

Q.17 [2 Marks] [MODERATE] [Theory of Machines — Cams]

A particle with constant velocity has acceleration:

A) Zero
B) Constant
C) Increasing
D) Decreasing

Q.18 [2 Marks] [MODERATE] [Heat Transfer — Convection]

Hooke's Law: stress proportional to:

A) Strain
B) Yield strength
C) Poisson's ratio
D) Density

Q.19 [2 Marks] [MODERATE] [Thermodynamics — Laws]

Reynolds number for laminar pipe flow:

A) Re < 1000
B) Re < 2000
C) Re < 4000
D) Re < 10000

Q.20 [2 Marks] [MODERATE] [Manufacturing — Welding]

Fourier's law negative sign indicates:

A) k is negative
B) Area negative
C) Heat flows opposite to temperature gradient
D) Heat flows along gradient

Q.21 [2 Marks] [MODERATE] [Fluid Mechanics — Flow]

First law of thermodynamics is based on:

A) Mass conservation

B) Energy conservation

C) Momentum conservation

D) Zeroth law

Q.22 [2 Marks] [MODERATE] [Engineering Mechanics — Friction]

In lathe, material is removed by:

A) Shear
B) Compression
C) Tension
D) Bending

Q.23 [2 Marks] [MODERATE] [Strength of Materials — Torsion]

Torque transmitted by shaft is proportional to:

A) D
B) D^2
C) D^3
D) D t

Q.24 [2 Marks] [MODERATE] [Theory of Machines — Governors]

Carnot cycle consists of:

A) 2 isothermal + 2 adiabatic
B) 4 isothermal
C) 4 adiabatic
D) 2 isothermal + 2 isobaric

Q.25 [2 Marks] [MODERATE] [Heat Transfer — Radiation]

A particle with constant velocity has acceleration:

A) Zero
B) Constant
C) Increasing
D) Decreasing

Q.26 [2 Marks] [MODERATE] [Thermodynamics — Laws]

Hooke's Law: stress proportional to:

A) Strain
B) Yield strength
C) Poisson's ratio
D) Density

Q.27 [2 Marks] [MODERATE] [Manufacturing — Forming]

Reynolds number for laminar pipe flow:

A) Re < 1000
B) Re < 2000
C) Re < 4000
D) Re < 10000

Q.28 [2 Marks] [MODERATE] [Fluid Mechanics — Fluid Properties]

Fourier's law negative sign indicates:

A) k is negative

B) Area negative

C) Heat flows opposite to temperature gradient

D) Heat flows along gradient

Q.29 [2 Marks] [MODERATE] [Engineering Mechanics — Kinematics]

First law of thermodynamics is based on:

A) Mass conservation
B) Energy conservation
C) Momentum conservation
D) Zeroth law

Q.30 [2 Marks] [MODERATE] [Strength of Materials — Stress-Strain]

In lathe, material is removed by:

A) Shear
B) Compression
C) Tension
D) Bending

Q.31 [2 Marks] [MODERATE] [Theory of Machines — Gears]

Torque transmitted by shaft is proportional to:

A) D
B) D^2
C) D^3
D) D t

Q.32 [2 Marks] [MODERATE] [Heat Transfer — Conduction]

Carnot cycle consists of:

A) 2 isothermal + 2 adiabatic
B) 4 isothermal
C) 4 adiabatic
D) 2 isothermal + 2 isobaric

Q.33 [2 Marks] [HARD] [Thermodynamics — Gas Cycles]

A particle with constant velocity has acceleration:

A) Zero
B) Constant
C) Increasing
D) Decreasing

Q.34 [2 Marks] [HARD] [Manufacturing — Forming]

Hooke's Law: stress proportional to:

A) Strain
B) Yield strength
C) Poisson's ratio
D) Density

Q.35 [2 Marks] [HARD] [Fluid Mechanics — Compressible Flow]

Reynolds number for laminar pipe flow:

A) $Re < 1000$

B) $Re < 2000$

C) $Re < 4000$

D) $Re < 10000$

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Engineering Mechanics — Friction]

Which are second law devices?

A) Heat engines
B) Heat pumps
C) Refrigerators
D) Isothermal processes
E) Adiabatic processes

Q.37 [2 Marks] [HARD] [Strength of Materials — Columns]

Which are conservative forces?

A) Gravity
B) Spring force
C) Friction
D) Air resistance
E) Electrostatic

Q.38 [2 Marks] [HARD] [Theory of Machines — Governors]

Which are second law devices?

A) Heat engines
B) Heat pumps
C) Refrigerators
D) Isothermal processes
E) Adiabatic processes

Q.39 [2 Marks] [HARD] [Heat Transfer — Conduction]

Which are conservative forces?

A) Gravity
B) Spring force
C) Friction
D) Air resistance
E) Electrostatic

Q.40 [2 Marks] [HARD] [Thermodynamics — Psychrometrics]

Which are second law devices?

A) Heat engines
B) Heat pumps
C) Refrigerators
D) Isothermal processes
E) Adiabatic processes

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Strength of Materials — Bending]
Carnot engine 600K!'300K. Max efficiency (%)?

Answer: ____________________

Q.52 [2 Marks] [HARD] [Theory of Machines — Flywheels]
Wall: A=10m^2, L=0.2m, k=0.5 W/mK, 9ECÓ# ²à Heat flow (W)?

Answer: ____________________

Q.53 [2 Marks] [HARD] [Heat Transfer — Boiling]
Shaft D=50mm, N=300rpm, <CÓC Õ a. Power transmitted (kW)?

Answer: ____________________

Q.54 [2 Marks] [HARD] [Thermodynamics — Laws]
Tensile test: 20mm dia, 60kN load. Stress (MPa)?

Answer: ____________________

Q.55 [2 Marks] [HARD] [Manufacturing — Machining]
Displacement from (0,0) to (3,4) = ?

Answer: ____________________

Q.56 [2 Marks] [HARD] [Fluid Mechanics — Boundary Layers]
Pipe d=100mm, Q=0.01 m^3/s. Velocity (m/s)?

Answer: ____________________

Q.57 [2 Marks] [HARD] [Engineering Mechanics — Friction]
Carnot engine 600K!'300K. Max efficiency (%)?

Answer: ____________________

Q.58 [2 Marks] [HARD] [Strength of Materials — Thin Cylinders]
Wall: A=10m^2, L=0.2m, k=0.5 W/mK, 9ECÓ# ²à Heat flow (W)?

Answer: ________________________

Q.59 [2 Marks] [HARD] [Theory of Machines — Flywheels]

Shaft D=50mm, N=300rpm, <CÓC Õ a. Power transmitted (kW)?

Answer: ____________________________

Q.60 [2 Marks] [HARD] [Heat Transfer — Boiling]

Tensile test: 20mm dia, 60kN load. Stress (MPa)?

Answer: ____________________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	A	**Q.2:**	A	**Q.3:**	B
Q.4:	C	**Q.5:**	B	**Q.6:**	A
Q.7:	C	**Q.8:**	A	**Q.9:**	A
Q.10:	A	**Q.11:**	B	**Q.12:**	C
Q.13:	B	**Q.14:**	A	**Q.15:**	C
Q.16:	A	**Q.17:**	A	**Q.18:**	A
Q.19:	B	**Q.20:**	C	**Q.21:**	B
Q.22:	A	**Q.23:**	C	**Q.24:**	A
Q.25:	A	**Q.26:**	A	**Q.27:**	B
Q.28:	C	**Q.29:**	B	**Q.30:**	A
Q.31:	C	**Q.32:**	A	**Q.33:**	A
Q.34:	A	**Q.35:**	B	**Q.36:**	ABC
Q.37:	ABE	**Q.38:**	ABC	**Q.39:**	ABE
Q.40:	ABC	**Q.41:**	ABE	**Q.42:**	ABC
Q.43:	ABE	**Q.44:**	ABC	**Q.45:**	ABE
Q.46:	ABC	**Q.47:**	ABE	**Q.48:**	ABC
Q.49:	ABE	**Q.50:**	ABC	**Q.51:**	50
Q.52:	500	**Q.53:**	7.85	**Q.54:**	191
Q.55:	5	**Q.56:**	1.27	**Q.57:**	50
Q.58:	500	**Q.59:**	7.85	**Q.60:**	191

Detailed Solutions

Question 1
[Engineering Mechanics — Kinetics]

A particle with constant velocity has acceleration:

Answer: (A)

a = dv/dt. If v constant, a = 0.

Question 2
[Strength of Materials — Bending]

Hooke's Law: stress proportional to:

Answer: (A)

<2 Ò Sμ. Stress proportional to strain within elastic limit.

Question 3
[Theory of Machines — Gears]

Reynolds number for laminar pipe flow:

Answer: (B)

Re < 2000: laminar. 2000-4000: transitional. >4000: turbulent.

Question 4
[Heat Transfer — Boiling]

Fourier's law negative sign indicates:

Answer: (C)

q = -k" T: heat flows from high to low temperature.

Question 5
[Thermodynamics — Entropy]

First law of thermodynamics is based on:

Answer: (B)

9ER Ò Ò W. Energy is conserved in any process.

Question 6
[Manufacturing — Forming]

In lathe, material is removed by:

Answer: (A)

Machining removes material by shear deformation at the tool-chip interface.

Question 7
[Fluid Mechanics — Compressible Flow]

Torque transmitted by shaft is proportional to:

Answer: (C)

T " D^3 for a shaft of diameter D under shear stress.

Question 8
[Engineering Mechanics — Impact]

Carnot cycle consists of:

Answer: (A)

Carnot: 2 isothermal + 2 reversible adiabatic processes.

Question 9
[Strength of Materials — Thin Cylinders]

A particle with constant velocity has acceleration:

Answer:

(A)

a = dv/dt. If v constant, a = 0.

Question 10 *[Theory of Machines — Flywheels]*

Hooke's Law: stress proportional to:

Answer: (A)

<2 Ò Sµ. Stress proportional to strain within elastic limit.

Question 11 *[Heat Transfer — Convection]*

Reynolds number for laminar pipe flow:

Answer: (B)

Re < 2000: laminar. 2000-4000: transitional. >4000: turbulent.

Question 12 *[Thermodynamics — Entropy]*

Fourier's law negative sign indicates:

Answer: (C)

q = -k" T: heat flows from high to low temperature.

Question 13 *[Manufacturing — Metrology]*

First law of thermodynamics is based on:

Answer: (B)

9ER Ò Ò W. Energy is conserved in any process.

Question 14 *[Fluid Mechanics — Fluid Properties]*

In lathe, material is removed by:

Answer: (A)

Machining removes material by shear deformation at the tool-chip interface.

Question 15 *[Engineering Mechanics — Impact]*

Torque transmitted by shaft is proportional to:

Answer: (C)

T " D³ for a shaft of diameter D under shear stress.

Question 16 *[Strength of Materials — Torsion]*

Carnot cycle consists of:

Answer: (A)

Carnot: 2 isothermal + 2 reversible adiabatic processes.

Question 17 *[Theory of Machines — Cams]*

A particle with constant velocity has acceleration:

Answer: (A)

a = dv/dt. If v constant, a = 0.

Question 18 *[Heat Transfer — Convection]*

Hooke's Law: stress proportional to:

Answer:

(A)

<2 Ò Sμ. Stress proportional to strain within elastic limit.

Question 19 *[Thermodynamics — Laws]*

Reynolds number for laminar pipe flow:

Answer: (B)

Re < 2000: laminar. 2000-4000: transitional. >4000: turbulent.

Question 20 *[Manufacturing — Welding]*

Fourier's law negative sign indicates:

Answer: (C)

q = -k" T: heat flows from high to low temperature.

Question 21 *[Fluid Mechanics — Flow]*

First law of thermodynamics is based on:

Answer: (B)

9ER Ò Ò W. Energy is conserved in any process.

Question 22 *[Engineering Mechanics — Friction]*

In lathe, material is removed by:

Answer: (A)

Machining removes material by shear deformation at the tool-chip interface.

Question 23 *[Strength of Materials — Torsion]*

Torque transmitted by shaft is proportional to:

Answer: (C)

T " D³ for a shaft of diameter D under shear stress.

Question 24 *[Theory of Machines — Governors]*

Carnot cycle consists of:

Answer: (A)

Carnot: 2 isothermal + 2 reversible adiabatic processes.

Question 25 *[Heat Transfer — Radiation]*

A particle with constant velocity has acceleration:

Answer: (A)

a = dv/dt. If v constant, a = 0.

Question 26 *[Thermodynamics — Laws]*

Hooke's Law: stress proportional to:

Answer: (A)

<2 Ò Sμ. Stress proportional to strain within elastic limit.

Question 27 *[Manufacturing — Forming]*

Reynolds number for laminar pipe flow:

Answer:

(B)

Re < 2000: laminar. 2000-4000: transitional. >4000: turbulent.

Question 28 *[Fluid Mechanics — Fluid Properties]*

Fourier's law negative sign indicates:

Answer: (C)

q = -k" T: heat flows from high to low temperature.

Question 29 *[Engineering Mechanics — Kinematics]*

First law of thermodynamics is based on:

Answer: (B)

9ER Ò Ò W. Energy is conserved in any process.

Question 30 *[Strength of Materials — Stress-Strain]*

In lathe, material is removed by:

Answer: (A)

Machining removes material by shear deformation at the tool-chip interface.

Question 31 *[Theory of Machines — Gears]*

Torque transmitted by shaft is proportional to:

Answer: (C)

T " D^3 for a shaft of diameter D under shear stress.

Question 32 *[Heat Transfer — Conduction]*

Carnot cycle consists of:

Answer: (A)

Carnot: 2 isothermal + 2 reversible adiabatic processes.

Question 33 *[Thermodynamics — Gas Cycles]*

A particle with constant velocity has acceleration:

Answer: (A)

a = dv/dt. If v constant, a = 0.

Question 34 *[Manufacturing — Forming]*

Hooke's Law: stress proportional to:

Answer: (A)

<2 Ò Sμ. Stress proportional to strain within elastic limit.

Question 35 *[Fluid Mechanics — Compressible Flow]*

Reynolds number for laminar pipe flow:

Answer: (B)

Re < 2000: laminar. 2000-4000: transitional. >4000: turbulent.

Question 36 *[Engineering Mechanics — Friction]*

Which are second law devices?

Answer:

(A, B, C)

Heat engines, heat pumps, refrigerators are second law devices.

Question 37 *[Strength of Materials — Columns]*

Which are conservative forces?

Answer: (A, B, E)

Conservative: gravity, spring, electrostatic. Friction and drag are non-conservative.

Question 38 *[Theory of Machines — Governors]*

Which are second law devices?

Answer: (A, B, C)

Heat engines, heat pumps, refrigerators are second law devices.

Question 39 *[Heat Transfer — Conduction]*

Which are conservative forces?

Answer: (A, B, E)

Conservative: gravity, spring, electrostatic. Friction and drag are non-conservative.

Question 40 *[Thermodynamics — Psychrometrics]*

Which are second law devices?

Answer: (A, B, C)

Heat engines, heat pumps, refrigerators are second law devices.

Question 41 *[Manufacturing — Forming]*

Which are conservative forces?

Answer: (A, B, E)

Conservative: gravity, spring, electrostatic. Friction and drag are non-conservative.

Question 42 *[Fluid Mechanics — Compressible Flow]*

Which are second law devices?

Answer: (A, B, C)

Heat engines, heat pumps, refrigerators are second law devices.

Question 43 *[Engineering Mechanics — Work-Energy]*

Which are conservative forces?

Answer: (A, B, E)

Conservative: gravity, spring, electrostatic. Friction and drag are non-conservative.

Question 44 *[Strength of Materials — Columns]*

Which are second law devices?

Answer: (A, B, C)

Heat engines, heat pumps, refrigerators are second law devices.

Question 45 *[Theory of Machines — Cams]*

Which are conservative forces?

Answer:

(A, B, E)

Conservative: gravity, spring, electrostatic. Friction and drag are non-conservative.

Question 46 *[Heat Transfer — Radiation]*

Which are second law devices?

Answer: (A, B, C)

Heat engines, heat pumps, refrigerators are second law devices.

Question 47 *[Thermodynamics — Entropy]*

Which are conservative forces?

Answer: (A, B, E)

Conservative: gravity, spring, electrostatic. Friction and drag are non-conservative.

Question 48 *[Manufacturing — Casting]*

Which are second law devices?

Answer: (A, B, C)

Heat engines, heat pumps, refrigerators are second law devices.

Question 49 *[Fluid Mechanics — Pumps/Turbines]*

Which are conservative forces?

Answer: (A, B, E)

Conservative: gravity, spring, electrostatic. Friction and drag are non-conservative.

Question 50 *[Engineering Mechanics — Kinetics]*

Which are second law devices?

Answer: (A, B, C)

Heat engines, heat pumps, refrigerators are second law devices.

Question 51 *[Strength of Materials — Bending]*

Carnot engine 600K!'300K. Max efficiency (%)?

Answer: (50)

;r Ò   Ò T_c/T_h = 1 - 300/600 = 0.5 = 50%.

Question 52 *[Theory of Machines — Flywheels]*

Wall: A=10m^2, L=0.2m, k=0.5 W/mK, 9ECÓ# ^{2}à Heat flow (W)?

Answer: (500)

Q = kA9EBôÂ Ò ã]s s# ó ã" Ò S   W.

Question 53 *[Heat Transfer — Boiling]*

Shaft D=50mm, N=300rpm, <CÓC Õ a. Power transmitted (kW)?

Answer: (7.85)

P = (16/" 3) × <B r â r K2 –â &÷ W" Væ—G0...

Question 54 *[Thermodynamics — Laws]*

Tensile test: 20mm dia, 60kN load. Stress (MPa)?

Answer:

<2 Ò ô Ò c òƒÀ×0.01^2) = 191 MPa.

Question 55 *[Manufacturing — Machining]*

Displacement from (0,0) to (3,4) = ?

Answer: (5)

" $(3^2 + 4^2)$ = 5 units.

Question 56 *[Fluid Mechanics — Boundary Layers]*

Pipe d=100mm, Q=0.01 m^3/s. Velocity (m/s)?

Answer: (1.27)

v = Q/A = 0.01/(< s ã ["' $, ã#r Ò÷0.

Question 57 *[Engineering Mechanics — Friction]*

Carnot engine 600K!'300K. Max efficiency (%)?

Answer: (50)

;r Ò Ò T_c/T_h = 1 - 300/600 = 0.5 = 50%.

Question 58 *[Strength of Materials — Thin Cylinders]*

Wall: A=10m^2, L=0.2m, k=0.5 W/mK, 9ECÓ# ²à Heat flow (W)?

Answer: (500)

Q = kA9EBôÂ Ò ã]s s# ó ã" Ò S W.

Question 59 *[Theory of Machines — Flywheels]*

Shaft D=50mm, N=300rpm, <CÓC Õ a. Power transmitted (kW)?

Answer: (7.85)

P = (16/" 3) × <B r â r K2 –â &÷ W" Væ—G0...

Question 60 *[Heat Transfer — Boiling]*

Tensile test: 20mm dia, 60kN load. Stress (MPa)?

Answer: (191)

<2 Ò ô Ò c òƒÀ×0.01^2) = 191 MPa.
